“ON THE POLYNOMIAL VALUES REPRESENTED BY QUADRATIC FORMS”

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This paper solves the shortest (in terms of “length” as defined in the paper) up to now open Diophantine equation $y^2 + x^3y + z^2 + 1 = 0$, in the sense of showing that it has infinitely many solutions in integers. This is part of a systematic study initiated by the first author, which aims at shedding light on the boundary between decidable and undecidable Diophantine equations (it is well known that solubility of Diophantine equations in undecidable in general), by ordering all such equations in a natural way and trying to solve them in order.

The problem is reduced to showing that there are infinitely many integers x such that $x^6 - 4$ can be written as a sum of two squares. This problem is then considered in a more general form, replacing $x^6 - 4$ by a polynomial $P \in \mathbb{Z}[x]$, which is assumed to be either of degree 3 or 4 (lower degree cases being already known) or of the form $R(Q(x))$ with R of degree 3 and Q of degree 2. An elementary “tangent-line” approach (the idea goes back to Diophantus himself) reduces the problem to finding the integral points on a conic satisfying additional congruence conditions, which can be handled by established theory. This more general approach is then used to solve four of the equations of length 9 (which is, after the original equation has been solved, the smallest length occurring for open equations) in the same way. The authors also generalize their approach so that it can deal with arbitrary non-degenerate quadratic forms representing $P(x)$; they give two worked examples for this. The results have been formalized in the proof assistant Lean using Aristotle, which uses AI models.

The paper is well-written overall, but there are a few mostly minor points that should be dealt with; they are listed below. It fits well into the scope of IJNT, and so I recommend to **accept** it, once these points have been dealt with.

Structural comments

S1. The discussion of Section 2 should lead to formal statements, analogous to Propositions 4.1 and 4.2. Then the current Propositions 3.1 and 3.2 can be deduced as corollaries.

S2. You should add a short discussion of the relationship between the general theory in Section 4 and the sum-of-two-squares case in Section 2, pointing out that the multiplicativity plays a role. It would also make sense to mention simplifications that can be obtained in the general case when the form is multiplicative.

S3. Explain *how* condition (c) in the two algorithms can be checked.

S4. The Conclusion (Section 5) is not really necessary (it is likely an artifact of LLM assistance). Information in this section that is not yet mentioned in the introduction can be moved there.

S5. Table 1. The “l” columns do not carry information and can be left out (the caption already says that these equations have length 9). Are these equations listed in some specific order? It would make sense to specify it.

Minor comments

T1. Submission metadata. ‘Leicestershare’ → ‘Leicestershire’.
T2. Abstract, last line. ‘has’ → ‘have’.
T3. Proof of Prop. 3.1. ‘The smallest u’ → ‘The smallest positive u’.
T4. Section 5. ‘to a Pell-type auxiliary equation’—when $\deg P = 3$, it is actually linear in one variable (but see Comment 4 above).
T5. Reference [1] is missing an identifier (arXiv:2510.01346 [cs.AI]).
T6. References [5,15,22]. Proper names should be capitalized (Hardy, Littlewood, Brauer, Manin, Fermat).
T7. Reference [11]. The year field should simply be ‘2024’ instead of ‘[2024] ©2024’.

Lean formalization

L1. Please set this up as part of a Lean project according to standard templates, including a `formalization.yaml` file and a `comparator` setup containing the main results of the paper in its `Challenge.lean` file. (You can ask your favorite LLM to set this up for you.) The Lean file compiles with current Mathlib, but without surrounding information as to Lean and Mathlib versions, it will soon no longer do so.
L2. I’ve asked Claude to have a look at the Lean file. It mentions the following points (which I didn’t check myself).
- The four non-squareness facts ($a = 17\,006\,096$, 8320, 208, 80) are discharged with `native_decide`, which silently adds the axiom `Lean.ofReduceBool` (trust in the compiled evaluator). They are trivially replaceable by `decide` or `norm_num` using the bounds already stated in the paper ($4123^2 < a < 4124^2$, etc.); this should be done if the paper claims an axiom-free formalization.
- The Lean proof of Proposition 4.2 is different from, and arguably cleaner than, the one in the paper: instead of Gauss’s theorem for the conic (31) in (x, w), it proves a residue-controlled Pell theorem (`genPell_infinite_cong`: some power of the fundamental unit is $\equiv (1, 0) \pmod{N}$, so the iterates stay in a fixed residue class) and applies it after completing the square. The paper could mention this alternative argument; the Lean file also observes that $\Delta \neq 0$ is not needed for Proposition 4.2 itself.
- Proposition 4.4 is verified in Lean directly from the explicit polynomial families (by `ring`), not via the tangent construction; this is fine but should be stated.
- The docstrings cite “[9, Prop. 5.4]” for Gauss’s theorem, whereas in the paper this is [11]; please synchronize the references in the Lean file with the final bibliography.